

NPG, TRPO, PPO

Daniil Zelezetsky

Higher School of Economics

March 12, 2026

Let τ be a trajectory $(s_0, a_0, s_1, a_1, \dots)$

Recall:

$$J(\pi) = E_{\tau \sim \pi} \left[\sum_{t=0}^T \gamma^t r(s_t) \right] \quad - \text{goodness of the } \pi, \pi \text{ depends on } \theta$$

Policy gradient theorem:

$$\nabla_{\theta} J(\pi) \approx E_{(s,a) \sim \pi} \left[Q^{\pi}(s,a) \nabla_{\theta} \log \pi(a|s) \right]$$

Optimization

Suppose we want to optimize $F(\theta)$ by θ in local neighbourhood

We approximate F by $\hat{F}(\theta) \approx F(\theta_0) + \nabla F(\theta_0)^T(\theta - \theta_0)$

Minimizing $\hat{F}(\theta)$ is the same as minimizing $\nabla F(\theta_0)^T(d)$, $d = \theta - \theta_0$

We want to find d that

1) minimize $\nabla F(\theta_0)^T(d)$

2) $\boxed{d^T d} < \epsilon$

↑
Distance

Using Lagrange multipliers, we find that $d_{opt} \propto -\nabla F(\theta_0)$

Optimization

Now, let's measure distance using non-identical matrix K :

We want to find d that

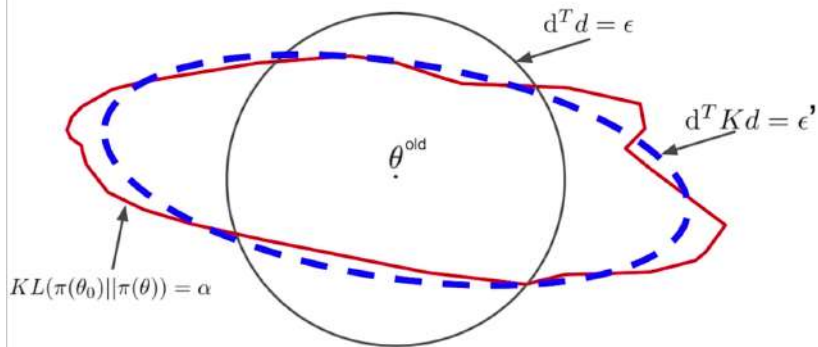
1) minimize $\nabla F(\theta_0)^T(d)$

2) $\boxed{d^T K d} < \epsilon$

↑
Distance

Using Lagrange multipliers, we find that $d_{opt} \propto -K^{-1}\nabla F(\theta_0)$

Space of parameters



Natural Policy Gradient

Suppose:

$$KL(\pi(\theta_0)||\pi(\theta)) \approx 0.5*(\theta-\theta_0)^T K(\theta-\theta_0) = 0.5*d^T K d, K = \nabla_{\theta}^2 KL(\pi(\theta_0)||\pi(\theta))$$

Solve constrained equation: find vector d that

1) Minimize $\nabla J^T d$

2) $d^T K d < \epsilon$

Solution: $d_{opt} \propto K^{-1} \nabla J(\theta)$

New update rule: $\theta_{t+1} = \theta_t - \alpha K^{-1} \nabla J(\theta)$

Problems: Value of learning rate still doesn't guarantee degree of policy change.

It may be too hard to compute inverse of K.

If we want to find $K^{-1}\nabla J(\theta)$ we may solve $Kx = \nabla J(\theta)$

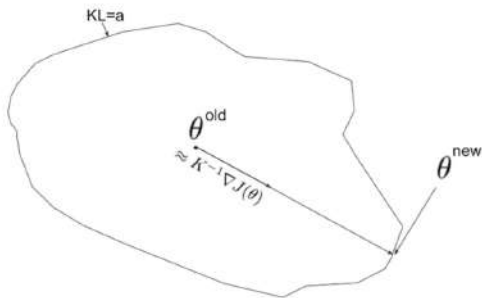
Matrix K is positive-definite so we can use **conjugate gradients**

Number of iterations k allows us to trade-off between precision and time.

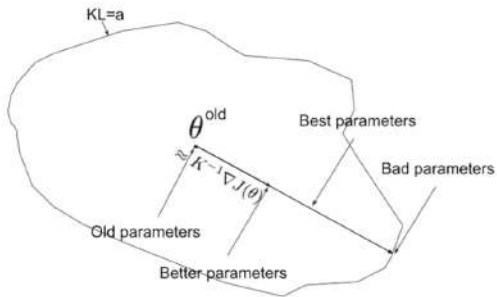
As a result: $\Delta\theta \approx \alpha K^{-1}\nabla J(\theta)$

Last problem: Value of learning rate still doesn't guarantee degree of policy change.

Let's do linear search! 16:21 / 59:17



Imagine this situation:



Recall: $J(\pi) = E_{\tau \sim \pi} \left[\sum_{t=0}^T \gamma^t r(s_t) \right]$

It can be proven that $J(\tilde{\pi}) = J(\pi) + E_{\tau \sim \tilde{\pi}} \left[\sum_{t=0}^T \gamma^t A_{\pi}(s_t, a_t) \right]$

Let's rewrite it this way $J(\tilde{\pi}) = J(\pi) + \sum_s \rho_{\tilde{\pi}}(s) \sum_a \tilde{\pi}(a|s) A_{\pi}(s, a)$

Trust Region trick:

If $E_s [KL(\pi || \tilde{\pi})]$ is small,

$$J(\tilde{\pi}) \approx J(\pi) + \sum_s \rho_{\pi}(s) \sum_a \tilde{\pi}(a|s) A_{\pi}(s, a)$$

Then:

$$\begin{aligned} J(\tilde{\pi}) &\approx J(\pi) + \sum_s \rho_{\pi}(s) \sum_a \tilde{\pi}(a|s) A_{\pi}(s, a) = \\ &= J(\pi) + \sum_s \rho_{\pi}(s) \sum_a \pi(a|s) * \frac{\tilde{\pi}(a|s)}{\pi(a|s)} * A_{\pi}(s, a) = \\ &= J(\pi) + \boxed{E_{(s,a) \sim \pi} \left[\frac{\tilde{\pi}(a|s)}{\pi(a|s)} A_{\pi}(s, a) \right]} \end{aligned}$$

Can be computed at every point!

Trust Region Policy Optimization

1) Sample state-action pairs from on-policy distribution

2) Compute $g = \nabla_{\theta'} \hat{J}(\tilde{\pi}) = \nabla_{\theta'} \frac{1}{N} \sum_{i=0}^N \frac{\tilde{\pi}(s_i, a_i)}{\pi(s_i, a_i)} A_{\pi}(s_i, a_i)$

$$K = \nabla_{\theta'}^2 \frac{1}{N} \sum_{i=0}^N KL(\pi(s_i) \parallel \tilde{\pi}(s_i))$$

3) Find $\hat{d} = -1 * \text{ConjGrad}(Kx = g)$

4) Do linear search in direction of $\hat{d}$, constraint $\frac{1}{N} \sum_{i=0}^N KL(\pi(s_i) \parallel \tilde{\pi}(s_i)) < \alpha$

and simultaneously check value of $\frac{1}{N} \sum_{i=0}^N \frac{\tilde{\pi}(s_i, a_i)}{\pi(s_i, a_i)} A_{\pi}(s_i, a_i)$

TRPO

Advantages

- Very stable training
- Good result

Disadvantages

- Cheap sampling is necessary
- Not easy to implement

KL constraint

Strict: TRPO

$$\begin{aligned} & \underset{\theta}{\text{maximize}} && \hat{\mathbb{E}}_t \left[\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_t \right] \\ & \text{subject to} && \hat{\mathbb{E}}_t [\text{KL}[\pi_{\theta_{\text{old}}}(\cdot | s_t), \pi_{\theta}(\cdot | s_t)]] \leq \delta \end{aligned}$$

Penalty: PPO-decay

$$\underset{\theta}{\text{maximize}} \hat{\mathbb{E}}_t \left[\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_t - \beta \text{KL}[\pi_{\theta_{\text{old}}}(\cdot | s_t), \pi_{\theta}(\cdot | s_t)] \right]$$

Different losses

Basic objective

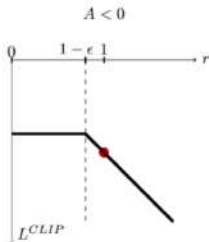
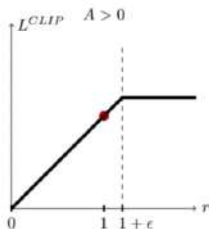
$$r_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, \text{ so } r(\theta_{\text{old}}) = 1$$

$$L^{CPI}(\theta) = \hat{\mathbb{E}}_t \left[\frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_t \right] = \hat{\mathbb{E}}_t \left[r_t(\theta) \hat{A}_t \right]$$

Different losses

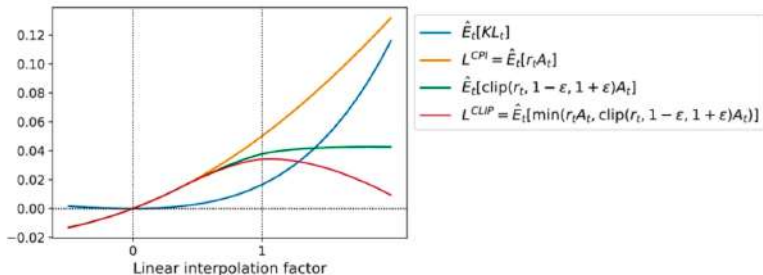
Clipped objective

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right]$$



Behaviour of losses

Suppose we interpolate the parameters of the model between updates



PPO-clip

Repeat:

- 1) Collect data
- 2) Update policy a few times to maximize $L^{CLIP}(\theta)$

algorithm	avg. normalized score
No clipping or penalty	-0.39
Clipping, $\epsilon = 0.1$	0.76
Clipping, $\epsilon = 0.2$	0.82
Clipping, $\epsilon = 0.3$	0.70
Fixed KL, $\beta = 0.3$	0.62
Fixed KL, $\beta = 1.$	0.71
Fixed KL, $\beta = 3.$	0.72
Fixed KL, $\beta = 10.$	0.69

Generalized Advantage Estimation (GAE)

Цель: получить хорошую оценку функции преимущества

$$A(s_t, a_t) = Q(s_t, a_t) - V(s_t)$$

Однако $Q(s_t, a_t)$ неизвестна. Вместо этого используем TD-ошибку:

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$$

Эта величина показывает, насколько фактическая награда отличается от предсказания value-функции. Generalized Advantage Estimation строится как экспоненциальная сумма TD-ошибок:

$$A_t = \delta_t + (\gamma\lambda)\delta_{t+1} + (\gamma\lambda)^2\delta_{t+2} + \dots$$

$$A_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}$$

где

$\gamma \in (0, 1)$ — discount factor

$\lambda \in [0, 1]$ — параметр сглаживания

Интуиция GAE

GAE можно интерпретировать как способ агрегировать TD-ошибки на разных временных горизонтах.

Если $\lambda = 0$:

$$A_t = \delta_t$$

мы используем только одношаговую TD-оценку.

Интуитивно GAE работает следующим образом:

- каждая TD-ошибка сообщает, насколько неожиданной оказалась награда
- ошибки распространяются вперёд во времени

В большинстве реализаций PPO используется

$$\gamma = 0.99, \quad \lambda = 0.95$$

Algorithm 1 PPO-Clip

- 1: Input: initial policy parameters θ_0 , initial value function parameters ϕ_0
- 2: **for** $k = 0, 1, 2, \dots$ **do**
- 3: Collect set of trajectories $\mathcal{D}_k = \{\tau_i\}$ by running policy $\pi_k = \pi(\theta_k)$ in the environment.
- 4: Compute rewards-to-go $\hat{R}_t$.
- 5: Compute advantage estimates, $\hat{A}_t$ (using any method of advantage estimation) based on the current value function V_{ϕ_k} .
- 6: Update the policy by maximizing the PPO-Clip objective:

$$\theta_{k+1} = \arg \max_{\theta} \frac{1}{|\mathcal{D}_k|T} \sum_{\tau \in \mathcal{D}_k} \sum_{t=0}^T \min \left(\frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_k}(a_t|s_t)} A^{\pi_{\theta_k}}(s_t, a_t), \quad g(\epsilon, A^{\pi_{\theta_k}}(s_t, a_t)) \right),$$

typically via stochastic gradient ascent with Adam.

- 7: Fit value function by regression on mean-squared error:

$$\phi_{k+1} = \arg \min_{\phi} \frac{1}{|\mathcal{D}_k|T} \sum_{\tau \in \mathcal{D}_k} \sum_{t=0}^T \left(V_{\phi}(s_t) - \hat{R}_t \right)^2,$$

typically via some gradient descent algorithm.

- 8: **end for**
-

TRPO

- Works for smaller models
- Second-order optimization

PPO

- Works for big models
- First-order optimization